

Defining causal effects

STSCI / INFO / ILRST 3900: Causal Inference

27 Aug 2026

Learning goals for today

By the end of class, you will be able to

- ▶ explain the fundamental problem of causal inference and the need for causal arguments
- ▶ define potential outcomes

Causal claims hinge on assumptions, not just data



¹Photo credit: YantImages via Wikipedia

Causal claims hinge on arguments, not just data

1. Statistical evidence

- ▶ Alysia Liu wore two skates in Milan. She won a gold medal.

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- ▶ Wearing two skates in Milan causes a person to win a gold medal.

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2. Possible causal claim

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What do we mean when we say “cause”?

Causal claims hinge on arguments, not just data

	Do you win gold if you:		Causal effect of wearing skates
	Wear skates	Do not wear skates	
Alysa Liu	Yes (1)	?	?
Sam	?	No (0)	?

Causal claims hinge on arguments, not just data

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Alysa Liu	Yes (1)	No (0)	+1
Sam	No (0)	No (0)	0

Observing



Image source: Wikimedia

Observation:

People who eat a Mediterranean diet have lower rates of cardiovascular disease

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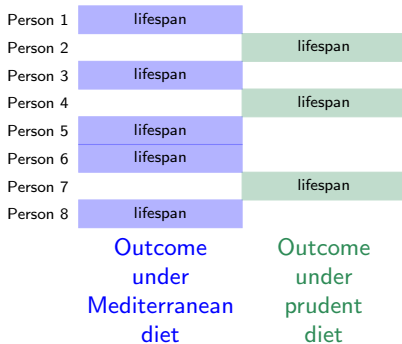
Discuss:

Do you think there is a causal effect here? If not, how might you explain away this apparent observation?

Fundamental problem of causal inference

Holland 1986

Descriptive evidence



Fundamental problem of causal inference

Holland 1986

Descriptive evidence



Causal claim



Person 1	lifespan	
Person 2		lifespan
Person 3	lifespan	
Person 4		lifespan
Person 5	lifespan	
Person 6	lifespan	
Person 7		lifespan
Person 8	lifespan	

Outcome
under
Mediterranean
diet

Outcome
under
prudent
diet

lifespan	lifespan
lifespan	lifespan
lifespan	lifespan
lifespan	lifespan
lifespan	lifespan
lifespan	lifespan
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Fundamental problem of causal inference

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Causal claim



Person 1	lifespan	missing
Person 2	missing	lifespan
Person 3	lifespan	missing
Person 4	missing	lifespan
Person 5	lifespan	missing
Person 6	lifespan	missing
Person 7	missing	lifespan
Person 8	lifespan	missing

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Causal claim



Causal inference is a **missing data** problem

Person 1	lifespan	missing
Person 2	missing	lifespan
Person 3	lifespan	missing
Person 4	missing	lifespan
Person 5	lifespan	missing
Person 6	lifespan	missing
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Mathematical notation: Potential outcomes²

²Capital letters and lowercase letters mean different things!

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Y_i Outcome

Whether person i survived

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Mathematical notation: Potential outcomes²

Y_i	Outcome	Whether person i survived
A_i	Treatment	Whether person i ate a Mediterranean diet

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Y_i^a	Potential Outcome	Outcome person i would realize if assigned to treatment value a

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Examples:

If assigned prudent diet	If assigned mediterranean diet
Died	Survived

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$A_{\text{Sam}} = \text{MedDiet}$ We observed that Sam ate a Mediterranean diet

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he would have survived

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$Y_{\text{Sam}}^{\text{PruDiet}} = \text{died}$ If Sam had been assigned a prudent diet
he would have died

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Practice

Using the slip of paper you received and the diet you follow, what is

- ▶ Y_i
- ▶ A_i
- ▶ Y_i^{MedDiet}
- ▶ Y_i^{PruDiet}

The consistency assumption

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- ▶ Consistency would not hold if “forcing” someone to eat a Mediterranean diet results in a different lifespan than what we observed for someone who chose the Mediterranean diet

The consistency assumption

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- ▶ Consistency implies that observed outcome for an individual who "chose" a specific treatment is same as what we would have observed if the individual was "assigned" that treatment
- ▶ Consistency would not hold if "forcing" someone to eat a Mediterranean diet results in a different lifespan than what we observed for someone who chose the Mediterranean diet
- ▶ Consistency implies treatments don't have hidden types
- ▶ Consistency would not hold if Italian olive oil resulted in a different lifespan than Greek olive oil

Practice: How would you say this in English?

We might wonder how a person's earnings relate to whether they hold a college degree

1. $E(\text{Earnings} \mid \text{Degree} = \text{TRUE}) > E(\text{Earnings} \mid \text{Degree} = \text{FALSE})$

2. $E(\text{Earnings}^{\text{Degree}=\text{TRUE}}) > E(\text{Earnings}^{\text{Degree}=\text{FALSE}})$

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► Average earnings are higher among those with college degrees

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- ▶ Average earnings are higher among those with college degrees

2. $E(\text{Earnings}^{\text{Degree}=\text{TRUE}}) > E(\text{Earnings}^{\text{Degree}=\text{FALSE}})$

- ▶ The average earning if everyone was assigned to get a degree is higher than the average earnings if everyone was assigned to not get a degree
- ▶ On average, getting a degree causes higher earnings

Practice:

1. On average, students who do the homework learn more than those who don't
2. On average, doing the homework causes more learning

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You can now

- ▶ Read Chapter 1 of [Hernán and Robins 2020](#)